

Swept-Source OCT Angiography Features of Melanotic Choroidal Tumors: An Analysis of 102 Consecutive Cases

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Purpose: To quantitatively analyze swept-source OCT angiography (SS-OCTA) features of choroidal melanoma and nevi, focusing on their distinguishing angiographic features.

Design: Case-series study.

Participants: This retrospective study included 102 consecutive patients with choroidal melanoma or nevi who were referred to Beijing Tongren Hospital between July 2019 and June 2024.

Main Outcome Measures: The choriocapillaris and choroid slabs were en face reconstructed. A 5-grade system was developed based on the density and morphology of disorganized vasculature in the choroid slab. Choriocapillaris flow deficit (CCFD%) void was calculated within circular regions defined by the tumor's radius and diameter.

Methods: This retrospective study included 102 consecutive patients with choroidal melanoma or nevi referred to Beijing Tongren Hospital between July 2019 and June 2024. The choriocapillaris and choroid slabs were en face reconstructed. A 5-grade system was developed based on the density and morphology of disorganized vasculature in the choroid slab. Choriocapillaris flow deficit (CCFD%) void was calculated within circular regions defined by the tumor's radius and diameter.

Results: Of all the 102 patients, 87 (85%) patients with qualified SS-OCTA images, comprising 61 cases of choroidal melanoma and 26 cases of nevi. Sixty melanoma cases (98%) exhibited disorganized tumor vasculature forming twisted vessel loops and networks on the choroid slab (grade ≥ 2). Nevi showed 2 types of intrinsic tumor vasculature patterns in the choroid slab: well-organized (grade 0 or 1) and melanoma-like disorganized networks (grade ≥ 2). Classified based on multimodal imaging risk score, nevi with a high risk for growth (≥ 2 risk factors) were more likely to exhibit melanoma-like disorganized networks tumor vasculature (82% vs. 27%, $P = 0.015$) than low-risk nevi. Analysis of the choriocapillaris slab revealed a higher prevalence of patchy avascular areas in melanomas compared to nevi. The CCFD% was significantly higher in choroidal melanomas than in nevi within circular regions defined by the tumor's radius (56.4% vs. 34.4%, $P < 0.001$) and diameter (44.9% vs. 23.4%, $P < 0.001$).

Conclusions: Swept-source OCT angiography enables detailed noninvasive assessment of intrinsic vasculature in melanotic choroidal tumors.

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Supplemental material available at www.ophtalmologyscience.org.

Choroidal tumors are vision- and life-threatening conditions. The growth of most choroidal tumors depends on abnormal angiogenesis,^{1,2} a process extensively studied through vascular imaging modalities. Genetic aberrations underlie abnormal angiogenesis in choroidal tumors. Histopathologic studies have revealed that the abnormal angiogenesis within the tumor, such as loops and vascular networks, are associated with the progression and metastases of choroidal melanoma.^{3–5} In GNA11/GNAQ-mutant choroidal melanoma cell lines, upregulated VEGF disrupts endothelial barrier integrity and promotes uveal

melanoma cell transendothelial migration.⁶ Furthermore, monosomy 3 is strongly associated with distant metastases and poor survival outcomes.^{7,8} Loss of chromosome 3 downregulates Von Hippel–Lindau expression, leading to hypoxia-inducible factor-2 alpha activation and subsequent pathological angiogenesis.⁹ Notably, GNA11/GNAQ mutations also link to capillary malformations in choroidal hemangiomas.^{10,11} Therefore, analyzing intrinsic vasculature provides not only clinical diagnostic value but also elucidates molecular mechanisms of tumor-associated angiogenesis.

Over recent decades, indocyanine green (ICG) angiography has been widely adopted to delineate tumor vasculature among various choroidal tumors, including melanoma, metastases, hemangiomas, and other lesions.^{12–14} As a key component of multimodal imaging, ICG angiography facilitates diagnosis, treatment planning, and longitudinal monitoring of choroidal tumors.¹⁵ However, its limited spatial resolution and dye leakage significantly hinder detailed visualization of intrinsic tumor vasculature. These limitations are especially pronounced in small melanocytic choroidal lesions, in which ICG angiography often shows diffuse hypofluorescence and thus fails to differentiate large nevi from early-stage choroidal melanomas.¹⁶

OCT angiography (OCTA) has emerged as a noninvasive, dye-free alternative for high-resolution imaging of tumor vasculature. Compared to spectral-domain OCT, swept-source OCTA (SS-OCTA) offers faster scanning speeds, improved axial resolution, and deeper tissue penetration.¹⁷ The longer wavelength of the laser source allows for better penetration of melanin in the retinal pigment epithelium and melanotic tumor, thereby enhancing visualization of choroidal lesions. Previous studies have qualitatively described OCTA vascular patterns in choroidal melanoma and nevi,^{18–24} quantitative comparisons between these entities remain limited.

In the present study, we hypothesize that the SS-OCTA, with its superior imaging resolution, can provide detailed noninvasive visualization of intrinsic tumor vasculature and offer diagnostic and prognostic insights, particularly in differentiating malignant melanomas from benign nevi and identifying high-risk nevi prone to growth. To test this hypothesis, we conducted a retrospective analysis of SS-OCTA features in choroidal melanoma and nevi, and developed a novel grading system based on full-thickness choroidal en face reconstructions to quantify and classify their distinguishing angiographic characteristics.

Methods

Ethics, Consent, and Permissions

The Medical Ethics Committee of Beijing Tongren Hospital approved the study protocol (TRECKY2018-056), and written informed consent was obtained from all participants. This study adheres to the tenets of the Declaration of Helsinki.

Identification, Screening, and Inclusion of Patients

This retrospective study analyzed 115 consecutive patients with choroidal tumors who were referred to Beijing Tongren Hospital between July 2019 and June 2024. The inclusion criteria were: (1) treatment-naïve choroidal tumors that were diagnosed as choroidal melanoma or nevi; (2) received SS-OCTA examination during the initial visit. The exclusion criteria were as follows: (1) formerly treated by plaque radiotherapy, laser photocoagulation, transpupillary thermotherapy, or intravitreal anti-VEGF ($n = 9$); (2) severe optic media opacities because of vitreous hemorrhage or cataract ($n = 7$), as indicated by SS-OCTA image quality score $< 3/10$. In all, 102 patients were included in the present study.

Clinical Diagnosis

All patients received thorough ocular evaluation by a senior ocular oncology expert (W.B.W.), including binocular indirect ophthalmoscope, fundus photography, autofluorescence (CLARUS 500, Zeiss), SS-OCT/OCTA (SVision Imaging, Ltd.), and other multimodal imaging examinations when necessary. Tumor thickness and largest tumor diameter were assessed by B-scan ultrasonography. The following clinical data were obtained at the first visit: gender, age, ocular and systemic medical history, visual acuity, intraocular pressure, and tumor configuration.

Patients suspected to having choroidal melanoma underwent orbital magnetic resonance imaging and ICG angiography.¹⁶ The diagnosis of choroidal melanoma was based on fundus features (eg, pigmented tumor, exudative retinal detachment), ultrasound appearances (mushroom or dome-shaped tumor with acoustic hollowness), and magnetic resonance imaging (hyperintense on T1-weighted, hypointense on T2-weighted images, and bright enhancement with gadolinium). Indocyanine green angiography typically demonstrated diffusely hypofluorescence in small melanoma, whereas hyperfluorescent in large melanoma. It primarily used to differentiate choroidal melanoma from other choroidal tumors, especially choroidal hemangioma.¹⁶ In contrast, choroidal nevi tend to have clearly defined margins and a flat or slightly elevated configuration and remain stable in size.²⁵ All choroidal nevi were monitored for at least 12 months to exclude documented growth, which is defined as an increase of ≥ 0.5 mm in basal diameter or thickness.²⁶

Choroidal nevi were stratified according to their risk of growth, defined as enlargement in basal dimension or thickness by ≥ 0.5 mm. Risk assessment was performed using the TFSOM-DIM score, which includes the following factors: thickness > 2 mm by ultrasound, subretinal fluid by OCT, symptoms of vision loss $\leq 20/50$ Snellen, orange pigment by autofluorescence, acoustic hollowness by ultrasound, and tumor largest basal diameter > 5 mm by fundus photography.²⁷ Choroidal nevi that have ≥ 2 risk factors for growth were further classified as high-risk nevi.²⁸

SS-OCTA Image Acquisition and Feature Extraction

The SS-OCTA device employed in this study was VG200G, which operated at 200 000 Hz at a 1050-nm wavelength. Axial resolution and lateral resolution were 3.8 μm and 10 μm , respectively. Firstly, all patients achieved pupil dilation. Next, guided by ultra-widefield fundus images, patients were asked to fixate on an internal target to optimize scan centration. The 6 mm $\times$ 6 mm, 9 mm $\times$ 9 mm, and 12 mm $\times$ 12 mm scan pattern was selected to cover the whole lesion as much as possible. These scan patterns consisted of raster scan protocols of 512, 768, and 1024 horizontal B-scans that covered an area of 6 mm $\times$ 6 mm, 9 mm $\times$ 9 mm, and 12 mm $\times$ 12 mm. The B-scans, which contained 512 A-scans each, were repeated 3 times. While scanning the tumor, the refractive module in the SS-OCTA device was adjusted to focus on the tumor's apex. Built-in eye-tracking software was used to minimize motion artifacts.

After imaging acquisition and automatic segmentation, Bruch membrane (BM) was manually delineated on all scans to correct segmentation errors caused by lesion elevation.¹⁸ If the BM cannot be visualized, the lower boundary of retinal pigment epithelium was used as a surrogate. Two slabs were analyzed to visualize intrinsic tumor vasculature. The choriocapillaris slab was defined as a 20 μm -layer immediately posterior to BM, whereas the choroid slab extended from BM to the sclera, encompassing the

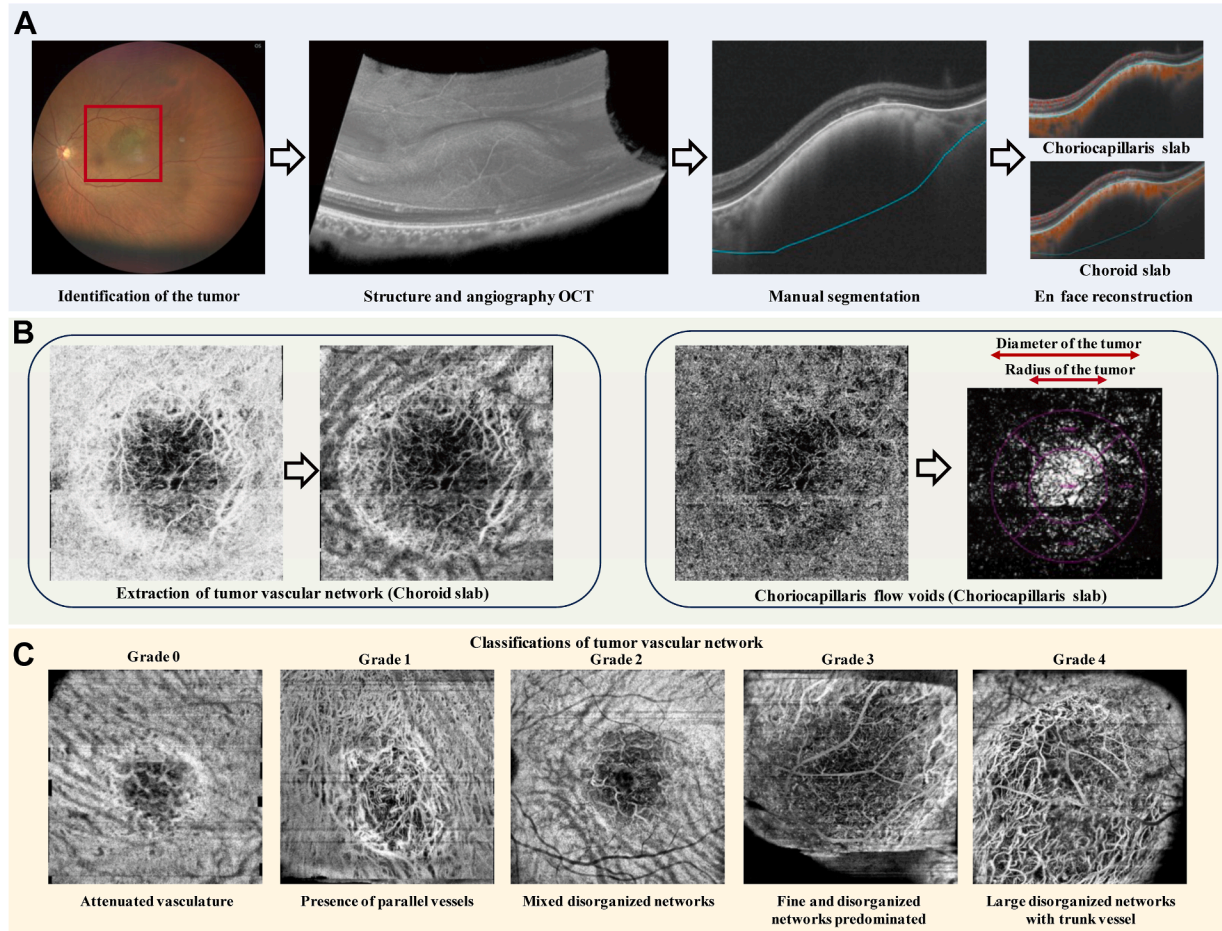


Figure 1. Diagram for SS-OCTA data acquisition, process, grading, and quantitative analysis. Guided by fundus images, structural and angiographic OCT scans were acquired and manually segmented. Red square indicates the OCTA scanning area. (A). The choroid slab and choriocapillaris slab were analyzed. The images contrast of choroid slab was automatically adjusted to highlight tumor vasculature. For choriocapillaris slab, the choriocapillaris flow deficits were quantified within circular regions defined by the tumor radius and diameter (B). The choroid slab was further graded into 5 categories based on the density and morphology of disorganized vasculature (C). OCTA = OCT angiography; SS-OCTA = swept-source OCT angiography.

entire choroidal vasculature. En face vascular maps were generated by projecting total flow signals within each slab (Fig 1A). For the choriocapillaris slab, choriocapillaris flow deficits (CCFD%) were quantified as the percentage of flow-void areas within 2 concentric circular regions centered on the tumor apex—defined by the tumor radius and diameter (Fig 1B). For the choroid slab, image contrast was automatically adjusted, and vascular patterns were manually graded into 5 categories.

- (1) Grade 0: tumor vasculature was not visible, showing attenuated flow signals due to dense melanin shadowing.
- (2) Grade 1: uniform parallel arrangement of medium-caliber vessels was observed within near-normal choroidal vasculature.
- (3) Grade 2: disorganized vascular networks were present centrally, involving less than 50% of the tumor diameter, whereas the peripheral regions retained near-normal choroidal vasculature.
- (4) Grade 3: disorganized vascular networks occupied more

than 50% of the tumor diameter, but consisted mainly of fine irregular vessels without trunk vascular.

- (5) Grade 4: a large and highly disorganized vascular network was observed, including thickened, tortuous vessels and prominent vascular trunks.

Two independent graders (R.H.Z. and W.H.T.) assessed vascular features. Intergrader agreement for tumor vasculature grading was high (Cohen kappa = 0.8966; Table S1, available at www.ophtalmologyscience.org). Discrepancies were resolved through discussion and arbitration by senior ocular oncology specialists (Y.M.L. and W.B.W.).

Statistical Analysis

Statistical analysis was performed in Stata version 17.0 (StataCorp LLC) and R Statistical Software (version 4.3.1; R Foundation for Statistical Computing). Student *t*-test, Wilcoxon rank-sum, and chi-square were used to determine differences in cross-sectional characteristics between groups for normal distribution, asymmetry distribution, and categorical data. A 2-sided *P* value <0.05 was considered for statistical significance.

Table 1. Baseline Characteristics of Included Patients

	Melanoma n = 61	Nevi n = 26	P Value
Age at diagnosis (yr)	47 ± 14	48 ± 14	0.33
Sex			0.06
Male (%)	32 (52%)	8 (31%)	
Female (%)	29 (48%)	18 (69%)	
Visual acuity (LogMAR)	0.7 [0.30–1.10]	0.09 [0.09–0.30]	<0.001
IOP (mmHg)	15.8 ± 3.7	15.2 ± 3.0	0.47
Largest basal diameter (mm)	9.0 [6.9–11.1]	3.3 [2.1–4.0]	<0.001
Tumor thickness (mm)	3.3 [2.1–4.7]	1.0 [0.6–1.8]	<0.001
Tumor configuration			<0.001
Plateau	9 (15%)	24 (92%)	
Dome	52 (85%)	2 (8%)	
Tumor anterior surface			0.33
Regular	50 (82%)	24 (92%)	
Irregular	11 (18%)	2 (8%)	
Subretinal fluid			<0.001
No	5 (8%)	24 (92%)	
<3 mm from margin	13 (21%)	2 (8%)	
>3 mm from margin	43 (70%)	0 (0%)	
Retinal edema			<0.001
No	33 (54%)	26 (100%)	
Yes	28 (46%)	0 (0%)	
Bruch membrane integrity			0.10
Normal	53 (87%)	26 (100%)	
Ruptured	8 (13%)	0 (0.0%)	

IOP = intraocular pressure; LogMAR = logarithm of the minimum angle of resolution.

Data are expressed as mean ± standard deviation, or median (interquartile range).

Results

Baseline Characteristics of Included Patients

Of the 102 treatment-naïve patients who underwent SS-OCTA examination, qualified images could not be acquired in 4 (5%) patients because of poor fixation, 5 (7%) because of exceeding the refractive module range, and 6 (8%) because of peripheral lesion location (Table S2, available at www.opthalmologyscience.org). A total of 87 (85%) patients with qualified SS-OCT images were included in the final analysis, comprising 61 cases of choroidal melanoma and 26 cases of choroidal nevi. None of the included patients had binocular involvement. Baseline characteristics of included patients were exhibited in Table 1.

According to the eighth edition of American Joint Committee on Cancer classification of posterior uveal melanoma,²⁹ 33 cases (54%) were categorized as T1, 23 (38%) as T2, 4 (7%) as T3, and 1 (2%) as T4. Eleven (42%) out of the 26 choroidal nevi had at least 2 risk factors for growth and were classified as high-risk nevi (Table S3, available at www.opthalmologyscience.org). As expected, patients with choroidal melanoma had poorer visual acuity, had larger tumor basal diameters and thicknesses, and were more likely to present with subretinal fluid and retinal edema.

Table 2. Comparison of SS-OCT/OCTA Feature of Choroidal Melanoma and Nevi

	Melanoma n = 61	Nevi n = 26	P Value
SS-OCT (cross-sectional)			
Choriocapillaris			<0.001
Normal	0 (0%)	5 (19%)	
Compressed	2 (3%)	19 (73%)	
Complete loss	59 (97%)	2 (8%)	
En face OCTA			
Choriocapillaris slab			
Patchy avascular area	47 (77%)	12 (46%)	0.006
Visualization of vessels	39 (64%)	8 (31%)	0.004
CCFD% (radius)	56.4 ± 21.4	34.4 ± 15.9	<0.001
CCFD% (diameter)	44.9 ± 20.1	23.4 ± 6.3	<0.001
Choroid slab			<0.001
Grade 0	0 (0%)	11 (42%)	
Grade 1	1 (2%)	2 (8%)	
Grade 2	22 (36%)	12 (46%)	
Grade 3	15 (25%)	1 (4%)	
Grade 4	23 (38%)	0 (0%)	

CCFD% = choriocapillaris flow deficit (within 2 circular regions centered at the tumor apex, with radii equal to 0.5 [radius] and 1.0 times [diameter] the tumor's basal diameter); OCTA = OCT angiography.

Data are expressed as number of patients (percentage).

Choroidal Melanoma Exhibited Disorganized Vasculature

OCT angiography images were successfully obtained in 61 of 76 patients (80%) with melanoma, compared to patients with melanoma who did not successfully acquire SS-OCTA. Patients with OCTA have significantly smaller largest basal diameter (9.0 mm vs 10.9 mm, $P = 0.03$) and tumor thickness (3.3 mm vs. 5.3 mm, $P = 0.03$, Table S4, available at www.opthalmologyscience.org). On cross-sectional SS-OCT, 59 (97%) cases of melanoma exhibited loss of normal choriocapillaris structure (Table 2 and Fig 2B, F and J). Consistent with this choriocapillaris loss observed in the cross-sectional scans, 64% and 77% of choroidal melanoma exhibited disorganized tumor vessel and patchy avascular area on choriocapillaris slab (Fig 2D, H and L). Furthermore, the choroid slab further revealed intrinsic tumor vasculature of choroidal melanoma, which is typically characterized by a disorganized labyrinth of vessels lacking a conventional blood vessel hierarchy (Fig 2C, G and K).

These vessels were irregular in diameter, forming twisted loops and networks, which could be visualized in 60 (98%) cases of melanoma. Only 1 case (2%) of melanoma showed parallel vessels formation (grade 1). Further quantitative analysis revealed that 22 cases (36%), 15 cases (25%), and 23 cases (38%) of choroidal melanoma were classified into grades 2, 3, and 4, respectively, indicating that disorganized intratumoral vasculature was a common OCTA feature of choroidal melanoma (Table 2). Additionally, 3 cases (4.9%) of amelanotic choroidal melanoma were identified, and intrinsic tumor vasculature was also visualized (Fig 2I–L).

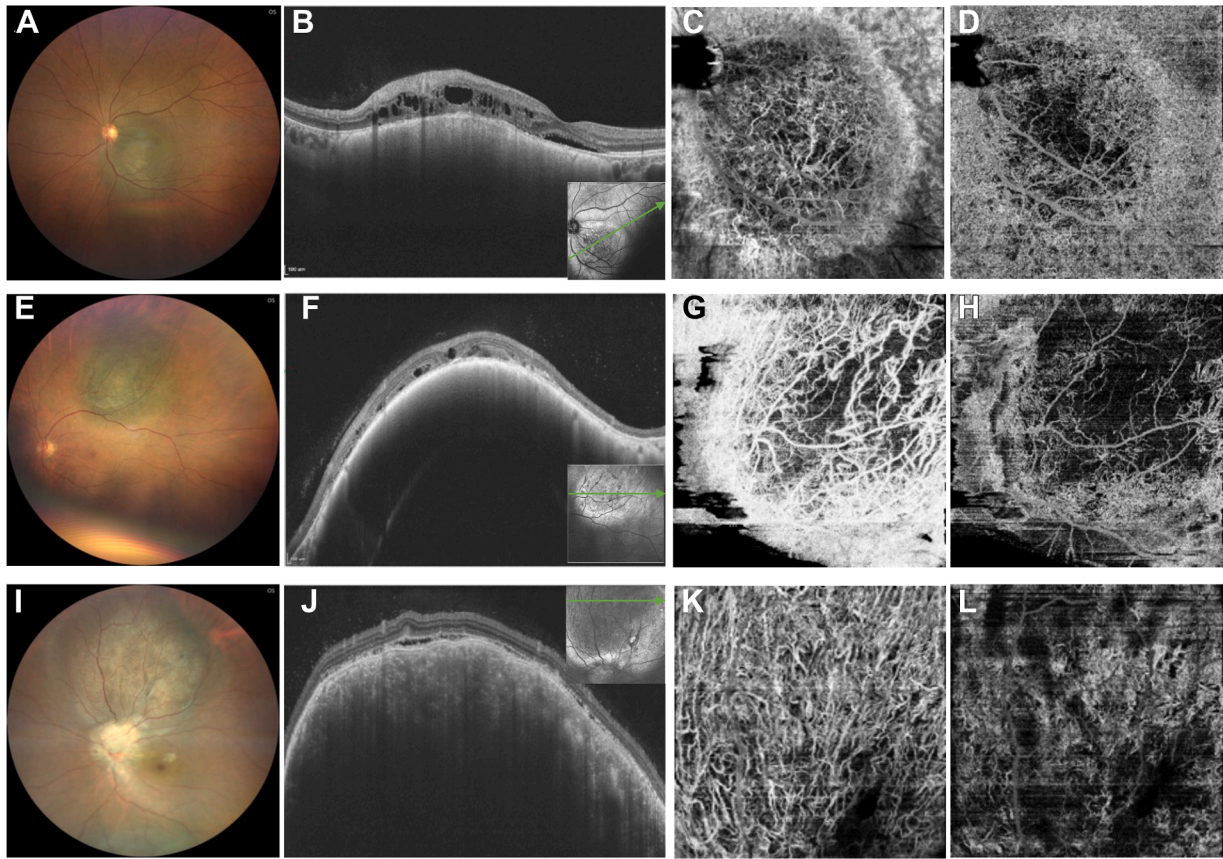


Figure 2. Swept-source OCTA images of choroidal melanoma. Three cases of choroidal melanoma showed typical intrinsic tumor vasculature. Fundus images showed tumor melanotic pigmentation in the first 2 cases (A, E) and amelanotic choroidal melanoma in the third case (I). A cross-sectional scan on the apex of the tumor showed choroidal mass with overall atrophied choriocapillaris. The green arrows indicate the scan line (B, F, J). En face OCTA reveals disorganized tumor vessels that formed into irregular vessel loops and networks (C: grade 3; G and K: grade 4). On choriocapillaris slab, the choriocapillaris structure was eroded by disorganized tumor vessels and patchy avascular area (D, H, L). OCTA = OCT angiography.

Choroidal Nevi Exhibited 2 Types of Vasculatures

OCT angiography images were successfully obtained in all 26 cases of nevi. On cross-sectional SS-OCT, the overlying choriocapillaris was compressed in 73% of nevi, in contrast to complete loss observed in 97% of choroidal melanoma cases ($P < 0.001$, Table 2). Compared to choroidal melanoma, a significantly smaller proportion of nevi exhibited patchy avascular area (77% vs 46%, $P = 0.006$) and tumor vasculature (64% vs 31%, $P = 0.004$) on choriocapillaris slab. The CCFD% was significantly higher in choroidal melanoma than nevi, both within the circular regions defined by the tumor radius ($56.4 \pm 21.4\%$ vs. $34.4 \pm 15.9\%$, $P < 0.001$) and diameter ($44.9 \pm 20.1\%$ vs. $23.4 \pm 6.3\%$, $P < 0.001$). Furthermore, the area under the curve of CCFD% for differentiating choroidal melanoma from choroidal nevi was 0.791 (95% confidence interval: 0.691–0.891) and 0.867 (0.791–0.941) within circular regions defined by the tumor radius and diameter, respectively ($P = 0.04$, Fig S1, available at www.ophtalmologyscience.org)

Two pattern of intrinsic tumor vasculature were identified in choroidal nevi. The first pattern was characterized

by attenuated blood vessel signaling, in which no twisted vessel loops or networks could be observed (Fig 3A–D). Quantitatively, this pattern of vascular pattern corresponds to grades 0 to 1 in the grading system (Fig 1C). The second pattern of vasculature harbored melanoma-like disorganized networks (Fig 3E–H), which harbors twisted vessel loops and networks. This pattern of vascular pattern corresponds to grades 2 to 4.

We further investigated the potential connection between melanoma-like disorganized networks and their multimodal imaging features. According to previous studies, a subgroup of choroidal nevi has been stratified as high risk for growth based on multimodal imaging.²⁷ A subsequent subgroup analysis was performed to compare of OCTA features between high-risk nevi and low-risk nevi. As shown in Table 3, although the presence of patchy avascular area was similar (64% vs. 33%, $P = 0.23$), nevi with high-risk for growth were more likely to exhibit detectable tumor vasculature on the choriocapillaris slab (55% vs. 13%, $P = 0.038$). Furthermore, a significantly higher proportion of high-risk nevi demonstrated disorganized vascular networks on the choroid slab and were more likely to have vasculature classified as grade ≥ 2 (82% vs. 27%, $P = 0.015$).

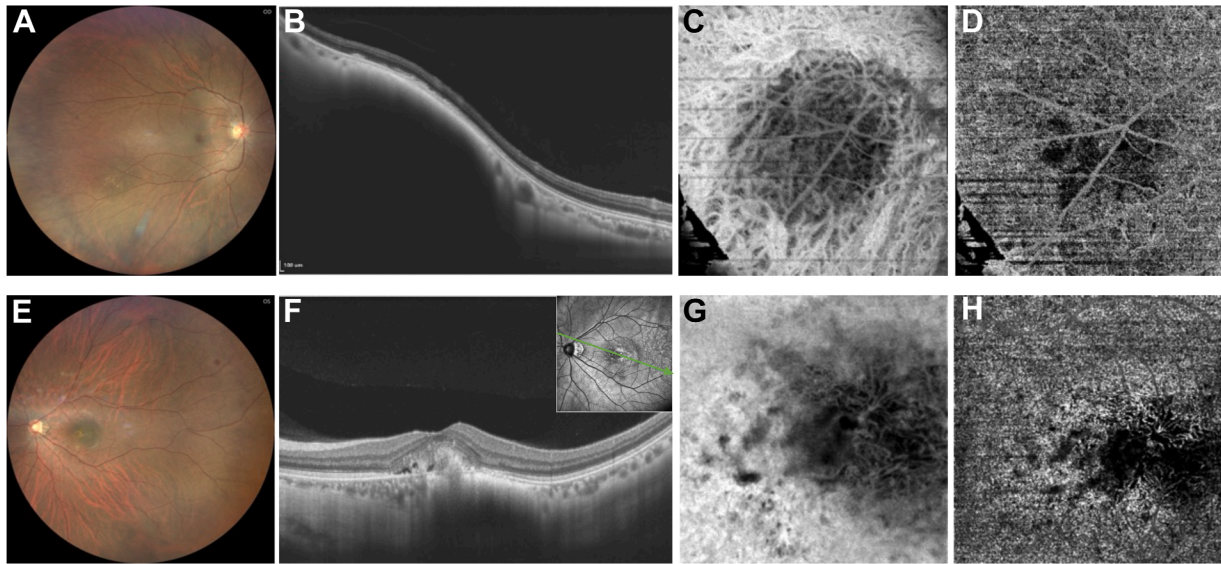


Figure 3. Swept-source OCTA images of choroidal nevi. Two types of intrinsic tumor vasculature in choroidal nevi: overall well-organized vasculature (A–D) and melanoma-like disorganized networks (E–H). Cross-sectional scan on the apex of tumor showed generally compressed choriocapillaris. The green arrows indicate the scan line (B, F). The below case of nevi has 3 risk factors: orange pigment by autofluorescence, acoustic hollowness by ultrasound, and subretinal fluid by OCT. The above case of nevi has one risk factor for malignant transformation: orange pigment by autofluorescence. Note that the melanoma-like disorganized networks harbored twisted vessel loops and networks (G, grade 3). OCTA = OCT angiography.

Discussion

This study focused on the OCTA features of choroidal melanotic tumors. For en face reconstruction of the full-thickness choroid, we developed a grading system based on the density and morphology of disorganized vasculature. Using this system, we found that choroidal melanomas

exhibited disorganized tumor vasculature, characterized by twisted vessel loops and networks. In contrast, choroidal nevi generally displayed attenuated or absent intralesional vascular signals. Notably, the presence of melanoma-like disorganized vascular networks within nevi was significantly associated with an increased risk of growth. For reconstruction of the choriocapillaris layer, we observed that patchy avascular areas and tumor vasculature were more frequently seen in choroidal melanomas than in nevi, which was associated with higher CCFD%.

Previous studies have investigated the OCTA features of choroidal melanoma.^{18–23,30,31} Pellegrini et al²⁰ reported that the vasculature of choroidal melanoma consists of both large vascular trunks and fine capillary-like networks. In the study by Pellegrini et al, the median tumor thickness of 4.0 mm and diameter of 10.5 mm were slightly larger than the present study. They found that about 27% ~ 36% of intrinsic vasculature could be missed by automatic segmentation. After manual segmentation, intrinsic vasculature could be visualized in all 22 cases of melanoma.²⁰ Another study found a similar vascular pattern in 12 cases of melanoma.¹⁹

We validated their findings in 60 (98%) cases of choroidal melanoma. Building upon these earlier descriptive studies, we further developed a five-grade classification system for tumor vasculature based on vessel density and morphological characteristics. According to this grading system, 98% of choroidal melanomas were classified as grade ≥ 2 , with only 1 case demonstrating a predominantly parallel vascular pattern (grade 1). These results suggest that aberrant tumor angiogenesis is a common feature of choroidal melanoma, regardless of American Joint Committee on Cancer stage. In smaller choroidal melanomas,

Table 3. Comparison of SS-OCT/OCTA Features of Low-Risk and High-Risk Nevi

	Low-Risk Nevi n = 15	High-Risk Nevi n = 11	P Value
SS-OCT (cross-sectional)			
Choriocapillaris			0.34
Normal	3 (20%)	2 (18%)	
Compressed	12 (80%)	7 (63%)	
Complete loss	0 (0%)	2 (18%)	
En face OCTA			
Choriocapillaris slab			
Patchy avascular area	5 (33%)	7 (64%)	0.23
Visualization of vessels	2 (13%)	6 (55%)	0.038
CCFD% (radius)	33.2 $\pm$ 14.1	36.1 $\pm$ 18.6	0.33
CCFD% (diameter)	23.6 $\pm$ 6.5	23.3 $\pm$ 6.2	0.45
Choroid slab			0.015
Grade 0	9 (60%)	2 (18%)	
Grade 1	2 (13%)	0 (0%)	
Grade 2	4 (27%)	8 (73%)	
Grade 3	0 (0%)	1 (9%)	
Grade 4	0 (0%)	0 (0%)	

CCFD% = choriocapillaris flow deficit; OCTA = OCT angiography. Data are expressed as number of patients (percentage).

disorganized vasculature on OCTA is typically confined to the central region of the tumor, corresponding to areas of greatest thickness. Toward the tumor margins, the choroidal vasculature often appeared relatively preserved (grade 2). As tumor size increased, the extent of vascular disorganization expanded (grade 3), eventually evolving into large disorganized networks with prominent trunk vessels (grade 4).

It remains challenging to distinguish between large choroidal nevi and early melanoma during a single clinical visit. To aid in early detection, considerable efforts have been made to predict the risk of malignant transformation of choroidal nevi. Shields et al proposed a multimodal imaging-based prediction model for choroidal nevi growth, using an increase of ≥ 0.5 mm in basal diameter or height as the endpoint. This model was successfully validated for risk stratification in a cohort of 3806 patients with choroidal nevi.²⁷ To explore the relationship between melanoma-like disorganized vascular networks and multimodal imaging features, we stratified choroidal nevi into 2 subgroups based on the Shields et al model: high-risk and low-risk for growth. We found that the presence of disorganized vascular networks (grade ≥ 2) within choroidal nevi was significantly more common in high-risk nevi than in low-risk nevi (82% vs. 27%). These observations indicated that the emergence of disorganized vascular networks may represent the histological basis underlying the growth potential of choroidal nevi. Therefore, choroidal nevi exhibiting grade ≥ 2 disorganized vascular patterns on OCTA should be closely monitored in clinical practice to identify early signs of malignant transformation.

It should be noted, however, that the TFSOM-DIM criteria used in this study are predictive of choroidal nevus growth in the future, and such growth does not necessarily indicate malignant transformation. In a fine needle biopsy study, 26 of 44 cases (59%) of nevi that did not show documented growth exhibited class 2 gene expression, a profile closely related to BRCA1 Associated Protein 1 mutated uveal melanoma.^{32,33} In contrast, 77 of 163 (37%) cases of nevi that showed documented growth exhibited class 1 gene expression, a profile closely related to benign melanotic lesion.³³ Most risk factors in TFSOM-DIM model, except for tumor thickness > 2.25 mm, were found to be not associated with class 2 gene expression in that study.³³ These findings suggest that the presence of melanoma-like disorganized vascular networks on SS-OCTA may not necessarily represent a risk of malignant transformation. Therefore, the potential correlation between SS-OCTA vascular abnormalities and molecular tumor class warrants further investigation. Such research would require long-term follow-up of a large cohort of patients with choroidal nevi.³⁴

Several researches explored the impact of delay treatment on the prognosis of uveal melanoma. Stålhammar et al found that the risk for uveal melanoma mortality increased by 1% for every additional 10-day delay in treatment. Interestingly, among 394 American Joint Committee on Cancer stage I patients in Stålhammar et al³⁵ study, cumulative incidence analysis showed no significant

difference in melanoma-specific mortality between those treated within 1 month of diagnosis and those with delayed treatment (≥ 1 month). Another study showed the tumor doubling time of choroidal nevi and small melanoma are 6392 and 717 days, which indicated that delaying definitive treatment for nevi and small melanoma increases the risk of metastatic death by nearly zero per month.³⁶ Therefore, for small, early-stage choroidal melanomas, a short delay in treatment does not compromise long-term prognosis. These findings further support the strategy of regular observation for large choroidal nevi until quantified growth is documented as the most appropriate clinical approach.

The present study has strengths, including a relatively large number of cases and advanced SS-OCTA device. However, several limitations should be acknowledged. First, visualization of intrinsic tumor vasculature required manual segmentation, especially regarding choroidal melanoma. The workload of segmentation might limit the adoption of this diagnostic technique. In the future, a fine-tuned automatic program is warranted to facilitate tumor segmentation. Second, all data were acquired using a single OCTA device at a single tertiary referral center. The generalizability of the proposed vascular grading system to other imaging platforms or patient populations remains uncertain. Third, the current study is limited to cross-sectional observations, and future longitudinal investigations are warranted to characterize the dynamic evolution of intrinsic vascular networks within these lesions. Fourth, choroidal tumors with large tumor thickness or located at peripheral choroid cannot be visualized. Fifth, manually distinguishing retinal vessel projection artifacts from intrinsic tumor vasculature was sometimes needed because of the projection artifacts. Despite of manual segmentation, the deep intrinsic vasculature in melanotic tumor cannot be visualized because of limited penetration of pigmented tissues. Sixth, tumor biopsy was not performed in cases included into the present study. Correlation between genetic aberration and manifest of intrinsic tumor vasculature need to investigate in the future.

Conclusion

In conclusion, SS-OCTA enables noninvasive visualization of tumor vasculature in small to medium-sized choroidal melanotic tumors. Disorganized tumor vasculature is commonly observed in choroidal melanoma and was also identified in choroidal nevi at high risk for growth. These vascular features may aid in earlier diagnosis, assist in treatment decision-making, and contribute to a better understanding of the mechanisms underlying abnormal angiogenesis in choroidal melanotic tumors.

Data Availability

The data and materials in this study are available from the corresponding author upon reasonable request.

Footnotes and Disclosures

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Abbreviations and Acronyms:

BM = Bruch membrane; **CCFD%** = choriocapillaris flow deficit; **ICG** = indocyanine green; **OCTA** = OCT angiography; **SS-OCTA** = swept-source OCT angiography.

Keywords:

Choroidal melanoma, Choroidalnevi, OCT angiography.

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